

3.1 Finding 1: Cross-Indicator Inconsistency

A foundational yet rarely scrutinized premise pervades the visual token pruning literature: that a single importance indicator can reliably separate indispensable tokens from expendable ones. Our investigation reveals that the indicators employed by existing methods naturally coalesce into two distinct families: *saliency*, which quantifies individual token importance (VisionZip [54], CDPruner [58]), and *diversity*, which promotes collective representativeness of the retained set (DivPrune [3], SCOPE [11]).

As visualized in Fig. 3, these indicators produce strikingly different token selections—even on the same image under the same compression ratio. VisionZip concentrates on regions receiving high CLS attention, typically semantically salient foreground objects, *irrespective of the user’s question*. CDPruner shifts the selection toward regions most relevant to the text query, while its DPP kernel simultaneously promotes geometric spread among the retained tokens. DivPrune, by contrast, distributes selections to maximize the minimum pairwise distance in the feature space, yielding broad spatial coverage at the cost of foreground detail. SCOPE occupies a middle ground: its submodular coverage term prevents token clustering in a narrow semantic region, while the CLS saliency weight steers selection toward informative patches.

This divergence arises because the four methods optimize along fundamentally *orthogonal indicator dimensions*:

- **Dimension A — Saliency** evaluates individual token importance and further decomposes into two sub-dimensions with distinct behaviors:
 - **A1 (Vision-Intrinsic):** “*Is this token intrinsically important in the image, regardless of the question?*”—relying solely on vision-encoder signals (e.g., CLS attention in VisionZip and SCOPE), producing a fixed ranking for a given image.
 - **A2 (Task Relevance):** “*Is this token relevant to the user’s current query?*”—requiring text–visual interaction (e.g., CDPruner computes cosine similarity between each visual token and the instruction embedding), yielding query-dependent rankings that differ across prompts.
- **Dimension B — Representativeness** evaluates whether the selected subset collectively represents the original token set, and further decomposes along two complementary spaces:
 - **B1 (Semantic Diversity):** “*Are the selected tokens semantically non-redundant in the embedding space?*”—enforced via set-theoretic objectives that penalize feature-space duplication (e.g., DivPrune’s MMDP maximizes the minimum pairwise distance; SCOPE’s submodular coverage maximizes the sum of nearest-neighbor similarities; CDPruner’s conditional DPP maximizes the volume spanned by the selected feature vectors).
 - **B2 (Spatial Coverage):** “*Does the selected set preserve at least one representative token from every spatial region of the image?*”—enforced via geometric partitioning of the token grid (e.g., Nuwa [39] divides the grid into $M \times M$ non-overlapping blocks and selects top- n tokens per block; HoloV [44] allocates per-crop budgets proportional to holistic scores). Notably, B1 and B2 are not redundant: a semantically diverse subset may still concentrate in a single spatial quadrant, while a spatially uniform selection may retain redundant tokens from texturally homogeneous regions.

Crucially, these dimensions are not merely orthogonal—their failure modes are *complementary*. Optimizing saliency alone (Dimension A) causes the retained tokens to cluster around a few high-attention foreground regions, leaving the semantic space severely under-covered; SCOPE’s analysis quantifies this: under pure CLS-attention selection on LLaVA-1.5, the θ -coverage of the retained set falls *below that of random sampling* across all threshold values [11]. Optimizing semantic diversity alone (B1) yields the opposite pathology: DivPrune distributes tokens broadly in the embedding space but is agnostic to which regions are task-relevant, retaining background noise at the expense of fine-grained foreground details. Furthermore, even a semantically diverse subset (B1) may remain spatially skewed—tokens from texturally varied but co-located patches can achieve high pairwise distance in feature space while leaving entire image quadrants unrepresented, a systematic weakness for OCR, spatial reasoning, and referring expression comprehension [39].

These complementary failure modes reveal that no single indicator dimension suffices: saliency without representativeness discards context; representativeness without saliency discards detail; and semantic diversity without spatial awareness discards regions. This analysis motivates the design of a unified pruning framework that jointly optimizes across all four sub-dimensions (A1, A2, B1, B2), as we present in Sec. 4.